SRIDHAR M

AI ENGINEER | PYTHON | GENAI & AGENTIC WORKFLOWS

+91 9498351685 • sridharstudy417@gmail.com • Bengaluru, India

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Engineer and software engineer with 2+ years of professional experience building AI-enabled applications, Python backend services, and interactive engineering software. Hands-on with LangGraph/LangChain workflows, tool integration, RAG, embeddings, local/open-weight model integration, and evaluation workflows. Strong foundation in FastAPI, REST APIs, asynchronous processing, testing, and performance optimization.

TECHNICAL SKILLS

AI & GenAI: LangGraph, LangChain, agentic workflows, tool calling, RAG, embeddings, vector search, LLM integration, evaluation
Python & Backend: Python, FastAPI, REST APIs, SQLAlchemy, WebSockets, asynchronous processing, API integration
Models & Data: Ollama, vLLM, Hugging Face, PyTorch, NumPy, Pandas, local/open-weight model workflows
Data Stores: PostgreSQL, SQLite, Redis, ChromaDB
Frontend & Visualization: React, TypeScript, JavaScript, Three.js, browser-based visualization
Engineering: Git, Docker, Linux, automated tests, data schemas, performance profiling, CI/CD familiarity

PROFESSIONAL EXPERIENCE

Zenteiq Aitech Innovations | **AI Engineer**
Bengaluru, India • Mar 2026 – Present

- Build AI-enabled engineering and simulation applications combining agentic workflows, physics/engineering models, and specialized data.
- Develop stateful coding-agent and workflow-orchestration components, including tool integration and evaluation workflows.
- Build Python/FastAPI services and numerical-processing pipelines for document intelligence, workflow automation, and engineering workloads.
- Integrate open-weight models and domain-specific data into application workflows, with emphasis on useful outputs and dependable behavior.

NovaCast India | **Associate Software Engineer**
Coimbatore, India • Jun 2024 – Feb 2026

- Built browser-based CFD visualization software for multi-gigabyte simulation datasets, connecting Python services with React/TypeScript interfaces.
- Reduced initial visualization latency by approximately 30% through progressive binary streaming over Socket.IO.
- Reduced peak server memory usage by approximately 40% by improving chunked simulation-file parsing and ingestion.
- Developed data schemas, indexing, asynchronous pipelines, and automated tests for engineering software.

SELECTED AI & APPLIED ENGINEERING PROJECTS

ZIQ-CODE / evolving-test | *Agentic Coding Workflow* • github.com/sridhar-mani/evolving-test

- Explored cross-session convention memory, feedback-informed candidate rules, isolated Git worktrees, pre-tool security hooks, and prompt layouts for local inference.
- 10-instance SWE-bench Lite pilot: 8/10 tasks completed versus 7/10 for a baseline; small sample, not statistically conclusive.

AI Document Analyzer | *RAG Application*

- Built a React/FastAPI application with document retrieval, embedding-based search, Socket.IO progress updates, and local LLM integration.
- Used ChromaDB and Hugging Face embeddings for document question-answering workflows.

CADFusion | *AI-Assisted CAD & Simulation Workspace*

- Integrated a LangChain ReAct agent with deterministic CAD tools and worker-backed geometry operations.
- Combined React/TypeScript, Three.js, OpenCascade, WebAssembly, and WebGPU workflows.

TodoAI | *Stateful Tool-Using Assistant*

- Prototyped multi-step task execution using LangGraph, tool calling, and Ollama-backed inference.

EDUCATION & CERTIFICATION

- B.Tech, Anna University, Chennai | 2019–2023
- IIT Madras Diploma in Programming | Oct 2024
- IIT Madras Diploma in Data Science | Sep 2025
- NVIDIA Certification: Building RAG Agents with LLMs